Practical 4 Topic: Mutability of Lists, Immutability of Tuples, Tuple Assignment, Variable-Length Arguments and Dictionary Operations

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Q1. Write a Python program to demonstrate that Lists are mutable by updating an existing element.

Code

```
# Demonstration of List Mutability

colors = ["Red", "Green", "Blue"]

print("Original List:", colors)

colors[1] = "Yellow"

print("Modified List:", colors)
```

Output

```
Original List: ['Red', 'Green', 'Blue']
Modified List: ['Red', 'Yellow', 'Blue']
```

Q2. Write a Python program to demonstrate that Tuples are immutable.

Code

```
# Demonstration of Tuple Immutability

student = ("Rahul", 20, "Mumbai")

print("Original Tuple:", student)

student[1] = 25
```

Output

```
Original Tuple: ('Rahul', 20, 'Mumbai')

TypeError:
```

```
'tuple' object does not support item assignment
```

Q3. Write a Python program to swap two numbers using Tuple Assignment.

Code

```
a = int(input("Enter First Number: "))
b = int(input("Enter Second Number: "))

print("Before Swapping")
print("a =", a)
print("b =", b)

a, b = b, a

print("After Swapping")
print("a =", a)
print("b =", b)
```

Q4. Write a Python program to unpack a Tuple into individual variables.

Code

```
student = ("Amit", 21, "Mumbai")

name, age, city = student

print("Name :", name)
print("Age :", age)
print("City :", city)
```

Q5. Write a Python program to calculate the sum of any number of values using Variable-Length Arguments (*args).

Code

```
def addition(*numbers):

    total = 0

    for i in numbers:
```

```
        total += i

    print("Sum =", total)

addition(10, 20)
addition(10, 20, 30)
addition(10, 20, 30, 40)
```

Q6. Write a Python program to create a Dictionary containing student information and display all key-value pairs.

Code

```
student = {
    "Roll":101,
    "Name":"Rahul",
    "Course":"B.Sc.IT",
    "Marks":88
}

for key, value in student.items():
    print(key, ":", value)
```

Q7. Write a Python program to update and delete elements from a Dictionary.

Code

```
employee = {
    "ID":101,
    "Name":"Amit",
    "Salary":35000
}

print("Original Dictionary")
print(employee)

employee["Salary"] = 45000

employee["Department"] = "IT"

del employee["ID"]

print("\nUpdated Dictionary")
print(employee)
```

Q8. Write a Python program to count the frequency of each character in a given string using a Dictionary.

Code

```
text = input("Enter a String: ")

frequency = {}

for ch in text:

    if ch in frequency:
        frequency[ch] += 1
    else:
        frequency[ch] = 1

print(frequency)
```

Q9. Write a Python program to merge two Dictionaries.

Code

```
student1 = {
    "Name":"Rahul",
    "Age":20
}

student2 = {
    "City":"Mumbai",
    "Marks":90
}

student1.update(student2)

print(student1)
```

Q10. Write a Python program to store the marks of five students in a Dictionary and display the topper.

Code

```
marks = {
    "Rahul":80,
    "Amit":92,
    "Riya":87,
    "Neha":95,
    "Soham":78
}

topper = max(marks, key=marks.get)

print("Topper :", topper)
print("Marks :", marks[topper])
```

Additional Practice Programs

Q11. Write a Python program to find the maximum and minimum elements in a List.

```
numbers = [25, 10, 80, 45, 67]

print("Maximum =", max(numbers))
print("Minimum =", min(numbers))
```

Q12. Write a Python program to remove duplicate elements from a List.

```
numbers = [10, 20, 30, 10, 20, 40, 50]

unique = []

for item in numbers:
    if item not in unique:
        unique.append(item)

print(unique)
```

Q13. Write a Python program to reverse a Tuple.

```
t = (10, 20, 30, 40, 50)

print(tuple(reversed(t)))
```

Q14. Write a Python program to sort a Dictionary by its keys.

```
student = {
    "B":85,
    "A":90,
    "D":70,
    "C":80
}

for key in sorted(student):
```

```
    print(key, ":", student[key])
```

Q15. Write a Python program to find whether a given key exists in a Dictionary.

```
student = {
    "Name":"Rahul",
    "Age":20,
    "City":"Mumbai"
}

key = input("Enter Key: ")

if key in student:
    print("Key Found")
else:
    print("Key Not Found")
```